

Question	Marking Guidance	Mark	Comments
05.1	1. (DNA) helicase; 2. (DNA) polymerase;	2	List Rule Applies Accept (DNA) ligase/ Primase/telomerase/ Topoisomerase/DNA gyrase Reject RNA Accept phonetic spellings
05.2	1. Changes tertiary structure of the enzyme; 2. (Enzyme) <u>active site</u> formed/able to be formed/ <u>active site</u> becomes complementary;	2	1. Accept tertiary symbol 3° 1. Ignore 3D 2. Reject refs to inhibition/inhibitors 2. Ignore refs to E-S complexes form 2. Ignore refs to substrate phosphorylation
05.3	(Phosphorylation/phosphate) makes substrates more reactive/raises their energy level(s)/lowers activation energy for the reaction;	1	Ignore provides energy unqualified Ignore refs to kinetic energy unqualified
05.4	1. ATM will not bind to (broken) DNA; 2. DNA not repaired/ cell still has broken DNA; 3. Cell division continues/tumour forms; 4. Tumour suppressor (gene) not effective/ not activated; 5. May have no effect in diploid/heterozygous (organism); 6. (Which) still has a functional ATM/ATM gene;	3 max	

